THE GLUEING FUNCTOR REFLECTS ISOMORPHISMS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 2375-2379 of sources/stacks-spaces-pushouts/pinned-original-source.tex; proof span is lines 2381-2390, the complete author proof (8 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 39 of queue_spaces-pushouts.json (pb=468, stmt 2375-2379, proof 2381-2390).

Editorial note. spaces-pushouts wave-1: fresh adjudication accept (2026-10-14); six-check dispatched pool (§2 row appended 02:4x).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0AF3: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 2319–2321. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0AF1: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 2272–2276. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0AEV: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 2046–2060. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 041Y: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 2390–2390. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *In Situation 0AEV the functor (0AF3) reflects isomorphisms.*

Proof. By a formal argument with base change, this reduces to the following question: A morphism $a : X' \to X$ of algebraic spaces such that $U \times_X X' \to U$ and $Y \times_X X' \to Y$ are isomorphisms, is an isomorphism. The family $\{U \to X, Y \to X\}$

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can be refined by an fpqc covering by Lemma 0AF1. Hence the result follows from Descent on Spaces, Lemma 041Y. □